



Figure 1: Graphs in Ganglia



Figure 3: Ganglia Web GUI

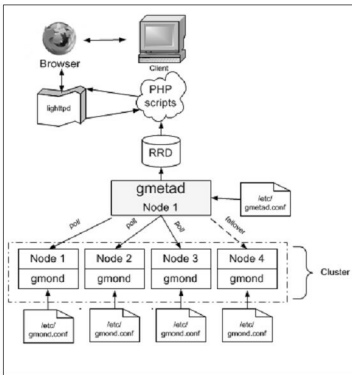


Figure 2: Ganglia architecture

```
/*
 * The cluster attributes specified will be used as part of the
 * <CLUSTER>
 * tag that will wrap all hosts collected by this instance.
 */
cluster {
  name = "mymachine"
  owner = "Konark"
  latlong = "unspecified"
  url = "unspecified"
}
```

Ganglia meta daemon

gmetad collects information from multiple *gmond* or *gmetad* sources. It saves the information to a local round-robin database, and exports (in XML) a concatenation of all the data sources. Below is a sample configuration:



Figure 4: Search

```
data_source "cluster name" [polling interval] address1:port
addresses2:port ...
```

The keyword '*data_source*' must immediately be followed by a unique string that identifies the source, and then by an optional polling interval in seconds. The source will be polled at this interval, on an average. If the polling interval is omitted, 15 seconds is assumed. A list of machines that service the data source follows, in the format *ip:port*, or *name:port*. If a port is not specified, 8649, the default *gmond* port, is assumed. In my case, when monitoring my local system, it is:

```
data_source "mymachine" 127.0.0.1
```

RRD

As defined earlier, RRDs are Round-Robin Databases. The data is archived after each hour—it is stored in an average form. You can demand the exact value of a particular instance of a particular month and hour. The defaults for RRDs are:

```
RRA:AVERAGE:0.5:1:244
RRA:AVERAGE:0.5:24:244
RRA:AVERAGE:0.5:168:244
RRA:AVERAGE:0.5:672:244
RRA:AVERAGE:0.5:5760:374
```

Ganglia PHP Web front-end

The front end presents the data collected by *gmond* and *gmetad* in a meaningful form. Ganglia has been around